

MERRYLAND HIGH SCHOOLS - ENTEBBE

S.4 BIOLOGY HOLIDAY REVISION ITEMS

SECTION A

Item 1

Judith, a senior three student fasted from morning till evening. Before breaking her fasting, she settled down for studies in the class room. A notorious boy, Ivan, a secret opium smoker, pelted a big stone which landed on the roof producing a loud sound. When Judith heard the loud sound, she ran out of the class very fast, breathing rapidly and her heart beat increased. This left other students wondering whether she fasted that day.

Task.

- (a) Describe how Judith's body coordinated from the time the loud sound was produced to the time she ran out of class.
- (b) What are the effects of Ivan's behaviour?
- (c) How can Ivan be helped to abandon that behaviour?

Item 2

On a certain island in Uganda, there are few health centers and as such, residents resorted to harvesting some plant parts and uprooting of plants in order to make herbal medicine. The practice worsened during the COVID -19 pandemic lockdown. This brought a threat of extinction of these medicinal plants on the island.

Task;

- (a) Identify the plant structures affected by the above practices.
- (b) Explain how the people's practices above affected the various processes in the plants
- (c) Suggest the control measures that can be done to overcome the extinction of the medicinal plants on the island.

Item 3

Nanyonga, a senior three student normally watches films at the nearby trading center up to midnight. On her way back home, she saw a fierce looking dog coming towards her. Immediately, she experienced an increase in the rate of heartbeat and breathing. She then turned back quickly and ran away.

Task

- (a) Describe how her body coordinated from the time she saw the dog up to when she ran away.
- (b) What are the possible effects of the student's way of life as described in the scenario above?
- (c) In which ways can the student be helped to change her lifestyle?

Item 4

Herbalists near Mabira forest use trees as the source of traditional medicines and many of those medicinal plant species are getting extinct. These herbalists cut roots, harvest leaves, debark trees and remove flowers from these trees.

Task.

- a) Identify the processes being affected by the practices of the traditional herbalists.
- b) Explain how the medicinal plants trees are able to survive despite the actions of herbalists.

Item 5

Mr. Butakabweggunda a resident in Mukono cleared all the vegetation on his flat piece of land and burnt it. He planted tomatoes since his neighbours have been gaining much profits from tomato growing in the past three years. In an effort to obtain maximum profits, he planted the tomato plants so close to one another and he never mulched his garden. Whenever it rained, the tomato leaves, stems and even flowers during flowering stage were found having a lot of soil particles on their surfaces. To his surprise, the yields never reached his expectations.

Task

- a) Explain the cause of the low yields from Mr. Butakabweggunda's garden
- b) Suggest what can be done by the farmer in order to have better harvests from his garden as his neighbours in the next season.

Item 6

Namutebi joined a peer group at school that was known for escaping at night from school to the neighboring trading center for cinemas, live concerts and clubbing. Their behaviour resulted into dismissal from school when the school administration was informed. By the time of expulsion, she was pregnant and she continued with her life style even after giving birth to her child that was hearing normally. After some time, she realized that her child was no longer responding to her

name and on diagnosis in the hospital, the baby's ear drum was found damaged. Also, Namutebi's health records obtained from school revealed that she had difficulty in copying notes from the chalkboard whenever she moved from her permanent seat on the second desk to the second last desk of her peer group friends in classroom.

Task

- a) Explain the cause of the challenges faced by Namutebi and her child.
- b) How can the challenges faced by Namutebi and her child be managed?

Item 7

During his first visit in January; Marvin a 2-year-old boy was found to be having wasted muscles, not physically active, had bleeding gums, pot belly, wounds on legs and was generally weak. The Doctor recommended that Marvin started a special diet mainly consisting of millet, milk, eggs and fruits for some time. Sarah, Marvin's mother was surprised that the Doctor recommended millet since she is a millet farmer.

Task:

- a) Describe the processes the millet in the diet will undergo in order to benefit Marvin
- b) Explain how the other components of the Doctor's recommendations in the diet will enable the child to recover.

Item 8

1. Tom is a famous tomato farmer in Kassanda. For better harvest, the tomatoes are regularly watered and fertilized, and the plants grow healthy with green leaves, large stems and large fruits. However, last season, soil fungus infected the plant's roots. This was followed by a sudden hail storm, which hit Tom's gardens damaging most parts of the plant. This resulted into very low harvest and poor-quality fruits.

Task

- a) Identify the plant parts that were affected resulting into low harvest
- b) Explain why the destroyed plant parts resulted into low plant yield?
- c) Suggest recommendations to Tom and other tomato farmers to overcome the occurrence of the same problem.

Item 9

A group of senior one students came to your class carrying a biological chart that was illustrating the different biogeochemical cycles that they couldn't understand the different components and processes in the chart. These students could only read the word title on one of their cycles which was "THE CARBON CYCLE" and they were wondering about the different processes and their roles in adding and removing carbon dioxide in the atmosphere. Some of these students were worried about the increased concentration of carbon dioxide that can result in global warming.

Task:

- a) Explain the different processes that can add and remove carbon dioxide in the atmosphere
- b) Suggest how high carbon dioxide levels can result in global warming
- c) Describe the likely effects of increased global warming in the atmosphere.

SECTION B

Item 10

In the state of environment Report, New Vision, May 21, 2023 reported that Kampala city generates over 1500 tons of waste every day and it is estimated that in urban areas, waste generation per person stands at around one kilogram per day, the NEMA Executive Director, Dr. Akankwasa Barirega said when contacted

Task

- (a) Identify the most common waste materials in the garbage from domestic sources.
- (b) What are the effects of the poor disposal of these materials.
- (c) Suggest what the community should do to properly manage waste in this area.

Item 11

In Namanve village, some farmers have settled in a section of the forest and wetland. The National Forest Authority, (NFA) and NEMA have issued eviction orders to the settlers from the forest and wetland. Most of the settlers are unhappy and threatened to resist the order.

Task

Sensitize the settlers on the environmental problems that the authorities want to address and explain why the village must conserve those resources.

Item 12

Anena is an athlete preparing for West Nile regional athletic competitions. He was confused about the meal to eat before the competition. Jamundu, the games teacher brought a meal of boiled cassava for Anena 3 hours before the competition. During the event, Anena outcompeted other athletes. He experienced, increase in heart rate, breathing rate and sweating.

Task

Identify the body systems that were involved from the time of the meal to the end of the race. Explain to Anena the sequence of processes undergone by boiled cassava in his body that caused him to win the race.

Item 13

Odong had been living with his uncle in Karamoja where it is relatively hot. He has joined senior one class at Lakweni high school. On reporting, he realized that it was a cold environment. He felt hungrier and his uncle took him to the nearby restaurant where the menu was either egg or rice but not both. His uncle recommended for rice but Odong was not happy because he wanted to eat eggs for breakfast. He also realized that the coldness also made him to urinate frequently, He became more worried.

Task;

Help Odong to understand what happens to the rice in his body to help him against the coldness, pointing out why he felt more hungrier and also urinated frequently yet he was not sick.

Item 14

Residents of a Bwaise, a developing town have pools of stagnant water near their houses. They use old pit latrines and pour their domestic wastes in the nearby garbage pits from where

they rot. Their children frequently fall sick and they visit nearby health centers. In this town, there are a few residents who planted some maize plants around their houses. However, it has been noted that these plants have been attacked by army worms (caterpillars).

Task.

(a) Explain to the residents how the conditions in their community caused the health state of their children.

(b) How can the effects in the scenario be minimized?

(c) Why should the pests mentioned be controlled?

Item 15

Soil is a non-renewable resource which should be used sustainably for continuity of life. In Busoga sub region-Eastern Uganda, people use soil for growing crops such as sugar canes and rice, mining of sand used for construction and plastering of buildings, as well as making of bricks. However, most people have limited knowledge on physical and chemical properties of soil, and as such, farmers get poor crop yields, plaster falls off walls of buildings, while bricks normally break after a very short period of time. This has caused serious economic losses.

Task;

- a) Explain the causes of each of the problems faced by the categories of people described in the scenario.
- b) Explain ways in which the problems faced by each category of people in a) above can be overcome.

Item 16

Yoweru, an athlete had a morning heavy carbohydrate meal before having a warm up in preparation for the marathon competition. He competed favourably, but towards the end, he developed sharp pains in his thigh muscles and was breathing deeply. Nonetheless, the athlete won the marathon. After finishing the race, his deep breathing continued, he sat down for one hour and the sharp high muscle pains progressively reduced. He later walked back home, though unable to understand all the experiences his body underwent.

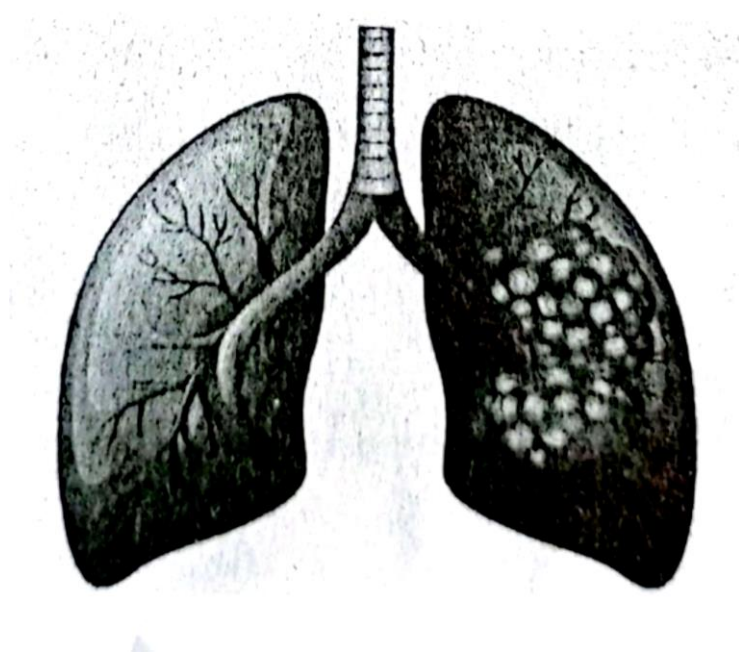
Task.

- a) Explain how Yoweri's body utilized the morning meal to enable him fully participate in the marathon
- b) Explain the difficulties his body met during the marathon and how the body overcame them to enable him walk back home.

Item 17

Mark, a student who has been in company of smokers for two years while at the vocational college developed dry cough, serious chest pain and was vomiting blood. His mother rushed him to hospital for thorough medical examination. The medical report indicated that he was a smoker, which he denied. The report further indicated that his blood could no longer carry enough

oxygen, and as a result, he was always very weak. He was then put on serious medication for management of his condition.



Task. a) Suggest the student's illness and explain the signs and symptoms mentioned in the scenario.

b) Advise the student on how he can overcome the problems associated with his health condition.

Item 18

James is a farmer growing maize on the slopes of a mountain. The mountain had thick vegetation cover on the mountain top before the volcanic eruption occurred. After sometime, James observed a large volume of water which flows rapidly from the mountaintop down the slopes, there are frequent landslides, the crop yield along the slopes has reduced, many animals that used to live on the mountain top have descended down the slopes and are destroying people's crops. Also people no longer get the medicinal plants from the mountain top and are suffering from most of the diseases they would treat before. Task

a) Explain the environmental challenges observed by James and how they can be addressed.

b) Why is it important to restore that environment back to its original state?

Item 19

John whose mother has hepatitis B was involved in a motor accident. He hit his head, lost a lot of blood and became unconscious. He was rushed to a nearby health center where he was discovered to be of blood group AB and had also injured his kidneys and lungs. The doctor advised that John should undergo blood transfusion in addition to other medication but the health center did not have blood compatible to his. Fortunately, John had been escorted by his friends by names of Eddy, Rogers, Simon, Esther, Patrick and Sharon whose blood compatibilities are: as follows.

Eddy can donate blood to Esther and Patrick, Rogers can donate blood to Simon and Sharon,

Esther can donate blood to Patrick, and Simon can donate blood to Sharon. Patrick can donate blood to Sharon but Sharon can't donate blood to Esther. Eddy has antibodies b in the blood plasma of his blood and his blood group is different from Patrick's. Simon has antibodies different from Eddy's and his blood group is different from that of Rogers'.

Task:

- a) Explain how the accident affected the functioning of John's body.
- b) Without referring John to another health center, suggest with reasons which friend of John can donate blood to him.

Item 20

Mr. Musisi had a piece of land with many shrubs and indigenous trees. He cleared the land in order to grow maize on large scale. In addition, he destroyed all ant hills with chemicals because he feared that termites would attack his maize. He also applied fertilisers in his maize garden. These caused many challenges to the environment.

Task: Explain;

- a) to Mr. Musisi the challenges he caused to the environment
- b) why it was important for Mr. Musisi to conserve his environment.

Item 21

A hilly place previously covered with thick vegetation has recently been dominated by many activities that include, timber-cutting, charcoal burning, rearing of cattle and crop gardening. Farmers practice bush burning in preparation for planting and also dig up and down the slopes of the hill, Incidences of strong wind, mud slides and flooding of gardens in the valleys have become common in the area.

Task: Explain;

- a) how the different activities have caused challenges in the area and how they can be overcome.
- b) why the community should conserve the hill.

Item 22

Mpaali is a good academician but also ambitious of becoming one of the best athletes. One day a competition was organized at school during which Mpaali and his friend Mbonye agreed to participate in cross country running. During 1 month prior to the day of competitions, Mbonye was always exercising and modifying his diet to include some rice, posho, beans, eggs and meat but Mpaali took these preparations for granted. In fact, Mbonye managed to win the race but Mpaali only managed to run three quarters of the journey.

Task:

- a) Describe how Mbonye's lifestyle prior to the day of competitions enabled him to win the race.
- b) Suggest reasons why Mbonye's managed to finish the race but Mpaali failed.

END

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